

Number Theory and Cryptography

Chapter 4

With Question/Answer Animations

Solving Congruences

Section 4.4

Section Summary₄

Linear Congruences

The Chinese Remainder Theorem

Computer Arithmetic with Large Integers (*not currently included in slides, see text*)

Fermat's Little Theorem

Pseudoprimes

Primitive Roots and Discrete Logarithms

Linear Congruences

Definition: A congruence of the form

$$ax \equiv b \pmod{m},$$

where m is a positive integer, a and b are integers, and x is a variable, is called a *linear congruence*.

The solutions to a linear congruence $ax \equiv b \pmod{m}$ are all integers x that satisfy the congruence.

Definition: An integer $\bar{a}$ such that $\bar{a}a \equiv 1 \pmod{m}$ is said to be an *inverse* of a modulo m .

Example: 5 is an inverse of 3 modulo 7 since $5 \cdot 3 = 15 \equiv 1 \pmod{7}$

One method of solving linear congruences makes use of an inverse $\bar{a}$, if it exists. Although we can not divide both sides of the congruence by a , we can multiply by $\bar{a}$ to solve for x .

Inverse of a modulo m

The following theorem guarantees that an inverse of a modulo m exists whenever a and m are relatively prime. Two integers a and b are relatively prime when $\gcd(a,b) = 1$.

Theorem 1: If a and m are relatively prime integers and $m > 1$, then an inverse of a modulo m exists. Furthermore, this inverse is unique modulo m . (This means that there is a unique positive integer $\bar{a}$ less than m that is an inverse of a modulo m and every other inverse of a modulo m is congruent to $\bar{a}$ modulo m .)

Proof: Since $\gcd(a,m) = 1$, by Theorem 6 of Section 4.3, there are integers s and t such that $sa + tm = 1$.

- Hence, $sa + tm \equiv 1 \pmod{m}$.
- Since $tm \equiv 0 \pmod{m}$, it follows that $sa \equiv 1 \pmod{m}$.
- Consequently, s is an inverse of a modulo m .
- The uniqueness of the inverse is Exercise 7.

THEOREM 6

BÉZOUT'S THEOREM If a and b are positive integers, then there exist integers s and t such that $\gcd(a, b) = sa + tb$.

Definition 6

If a and b are positive integers, then integers s and t such that $\gcd(a, b) = sa + tb$ are called *Bézout coefficients* of a and b (after Étienne Bézout, a French mathematician of the eighteenth century). Also, the equation $\gcd(a, b) = sa + tb$ is called *Bézout's identity*.

Finding Inverses₁

The Euclidean algorithm and Bézout coefficients gives us a systematic approaches to finding inverses.

Example: Find an inverse of 3 modulo 7.

Solution: Because $\gcd(3,7) = 1$, by Theorem 1, an inverse of 3 modulo 7 exists.

- Using the Euclidian algorithm: $7 = 2 \cdot 3 + 1$.
- From this equation, we get $-2 \cdot 3 + 1 \cdot 7 = 1$, and see that -2 and 1 are Bézout coefficients of 3 and 7.
- Hence, -2 is an inverse of 3 modulo 7.
- Also every integer congruent to -2 modulo 7 is an inverse of 3 modulo 7, i.e., 5, -9 , 12, etc.

Finding Inverses₂

Example: Find an inverse of 101 modulo 4620.

Solution: First use the Euclidian algorithm to show that $\gcd(101, 4620) = 1$.

$$4620 = 45 \cdot 101 + 75$$

$$101 = 1 \cdot 75 + 26$$

$$75 = 2 \cdot 26 + 23$$

$$26 = 1 \cdot 23 + 3$$

$$23 = 7 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$2 = 2 \cdot 1$$

Since the last nonzero remainder is 1, $\gcd(101, 4260) = 1$

Working Backwards:

$$1 = 3 - 1 \cdot 2$$

$$1 = 3 - 1 \cdot (23 - 7 \cdot 3) = -1 \cdot 23 + 8 \cdot 3$$

$$1 = -1 \cdot 23 + 8 \cdot (26 - 1 \cdot 23) = 8 \cdot 26 - 9 \cdot 23$$

$$1 = 8 \cdot 26 - 9 \cdot (75 - 2 \cdot 26) = 26 \cdot 26 - 9 \cdot 75$$

$$1 = 26 \cdot (101 - 1 \cdot 75) - 9 \cdot 75$$

$$= 26 \cdot 101 - 35 \cdot 75$$

$$1 = 26 \cdot 101 - 35 \cdot (4620 - 45 \cdot 101)$$

$$= -35 \cdot 4620 + 1601 \cdot 101$$

Bézout coefficients : - 35 and 1601

1601 is an inverse of 101 modulo 4620

Using Inverses to Solve Congruences

We can solve the congruence $ax \equiv b \pmod{m}$ by multiplying both sides by $\bar{a}$.

Example: What are the solutions of the congruence $3x \equiv 4 \pmod{7}$.

Solution: We found that -2 is an inverse of 3 modulo 7 (two slides back). We multiply both sides of the congruence by -2 giving

$$-2 \cdot 3x \equiv -2 \cdot 4 \pmod{7}.$$

Because $-6 \equiv 1 \pmod{7}$ and $-8 \equiv 6 \pmod{7}$, it follows that if x is a solution, then $x \equiv -8 \equiv 6 \pmod{7}$

We need to determine if every x with $x \equiv 6 \pmod{7}$ is a solution. Assume that $x \equiv 6 \pmod{7}$. By Theorem 5 of Section 4.1, it follows that $3x \equiv 3 \cdot 6 = 18 \equiv 4 \pmod{7}$ which shows that all such x satisfy the congruence.

The solutions are the integers x such that $x \equiv 6 \pmod{7}$, namely, $6, 13, 20$... and $-1, -8, -15, \dots$

Fermat's Little Theorem

Theorem 3: (*Fermat's Little Theorem*) If p is prime and a is an integer not divisible by p , then
 $a^{p-1} \equiv 1 \pmod{p}$

Furthermore, for every integer a we have
 $a^p \equiv a \pmod{p}$

(*proof outlined in Exercise 19*)

Fermat's little theorem is useful in computing the remainders modulo p of large powers of integers.

Example: Find $7^{222} \bmod 11$.

By Fermat's little theorem, we know that $7^{10} \equiv 1 \pmod{11}$, and so $(7^{10})^k \equiv 1 \pmod{11}$, for every positive integer k . Therefore,

$$7^{222} = 7^{22 \cdot 10 + 2} = (7^{10})^{22} 7^2 \equiv (1)^{22} \cdot 49 \equiv 5 \pmod{11}.$$

Hence, $7^{222} \bmod 11 = 5$.



Pierre de Fermat
(1601-1665)

Applications of Congruences

Section 4.5

Section Summary₅

Hashing Functions

Pseudorandom Numbers

Check Digits

Hashing Functions

Definition: A *hashing function* h assigns memory location $h(k)$ to the record that has k as its key.

- A common hashing function is $h(k) = k \bmod m$, where m is the number of memory locations.
- Because this hashing function is onto, all memory locations are possible.

Example: Let $h(k) = k \bmod 111$. This hashing function assigns the records of customers with social security numbers as keys to memory locations in the following manner:

$$h(064212848) = 064212848 \bmod 111 = 14$$

$$h(037149212) = 037149212 \bmod 111 = 65$$

$$h(107405723) = 107405723 \bmod 111 = 14, \text{ but since location 14 is already occupied, the record is assigned to the next available position, which is 15.}$$

The hashing function is not one-to-one as there are many more possible keys than memory locations. When more than one record is assigned to the same location, we say a *collision* occurs. Here a collision has been resolved by assigning the record to the first free location.

For collision resolution, we can use a *linear probing function*:

$$h(k, i) = (h(k) + i) \bmod m, \text{ where } i \text{ runs from } 0 \text{ to } m - 1.$$

There are many other methods of handling with collisions. You may cover these in a later CS course.

Pseudorandom Numbers₁

Randomly chosen numbers are needed for many purposes, including computer simulations.

Pseudorandom numbers are not truly random since they are generated by systematic methods.

The *linear congruential method* is one commonly used procedure for generating pseudorandom numbers.

Four integers are needed: the *modulus* m , the *multiplier* a , the *increment* c , and *seed* x_0 , with $2 \leq a < m$, $0 \leq c < m$, $0 \leq x_0 < m$.

We generate a sequence of pseudorandom numbers $\{x_n\}$, with $0 \leq x_n < m$ for all n , by successively using the recursively defined function
$$x_{n+1} = (ax_n + c) \bmod m.$$

(an example of a recursive definition, discussed in Section 5.3)

If pseudorandom numbers between 0 and 1 are needed, then the generated numbers are divided by the modulus, x_n/m .

Pseudorandom Numbers₂

Example: Find the sequence of pseudorandom numbers generated by the linear congruential method with modulus $m = 9$, multiplier $a = 7$, increment $c = 4$, and seed $x_0 = 3$.

Solution: Compute the terms of the sequence by successively using the congruence

$$x_{n+1} = (7x_n + 4) \bmod 9, \text{ with } x_0 = 3.$$

$$x_1 = 7x_0 + 4 \bmod 9 = 7 \cdot 3 + 4 \bmod 9 = 25 \bmod 9 = 7,$$

$$x_2 = 7x_1 + 4 \bmod 9 = 7 \cdot 7 + 4 \bmod 9 = 53 \bmod 9 = 8,$$

$$x_3 = 7x_2 + 4 \bmod 9 = 7 \cdot 8 + 4 \bmod 9 = 60 \bmod 9 = 6,$$

$$x_4 = 7x_3 + 4 \bmod 9 = 7 \cdot 6 + 4 \bmod 9 = 46 \bmod 9 = 1,$$

$$x_5 = 7x_4 + 4 \bmod 9 = 7 \cdot 1 + 4 \bmod 9 = 11 \bmod 9 = 2,$$

$$x_6 = 7x_5 + 4 \bmod 9 = 7 \cdot 2 + 4 \bmod 9 = 18 \bmod 9 = 0,$$

$$x_7 = 7x_6 + 4 \bmod 9 = 7 \cdot 0 + 4 \bmod 9 = 4 \bmod 9 = 4,$$

$$x_8 = 7x_7 + 4 \bmod 9 = 7 \cdot 4 + 4 \bmod 9 = 32 \bmod 9 = 5,$$

$$x_9 = 7x_8 + 4 \bmod 9 = 7 \cdot 5 + 4 \bmod 9 = 39 \bmod 9 = 3.$$

The sequence generated is 3,7,8,6,1,2,0,4,5,3,7,8,6,1,2,0,4,5,3,...

It repeats after generating 9 terms.

Commonly, computers use a linear congruential generator with increment $c = 0$. This is called a *pure multiplicative generator*. Such a generator with modulus $2^{31} - 1$ and multiplier $7^5 = 16,807$ generates $2^{31} - 2$ numbers before repeating.

Check Digits: UPCs

A common method of detecting errors in strings of digits is to add an extra digit at the end, which is evaluated using a function. If the final digit is not correct, then the string is assumed not to be correct.

Example: Retail products are identified by their *Universal Product Codes (UPCs)*. Usually these have 12 decimal digits, the last one being the check digit. The check digit is determined by the congruence:

$$3x_1 + x_2 + 3x_3 + x_4 + 3x_5 + x_6 + 3x_7 + x_8 + 3x_9 + x_{10} + 3x_{11} + x_{12} \equiv 0 \pmod{10}.$$

- a. Suppose that the first 11 digits of the UPC are 79357343104. What is the check digit?
- b. Is 041331021641 a valid UPC?

Solution:

- a. $3 \cdot 7 + 9 + 3 \cdot 3 + 5 + 3 \cdot 7 + 3 + 3 \cdot 4 + 3 + 3 \cdot 1 + 0 + 3 \cdot 4 + x_{12} \equiv 0 \pmod{10}$
 $21 + 9 + 9 + 5 + 21 + 3 + 12 + 3 + 3 + 0 + 12 + x_{12} \equiv 0 \pmod{10}$
 $98 + x_{12} \equiv 0 \pmod{10}$
 $x_{12} \equiv 2 \pmod{10}$ So, the check digit is 2.
- b. $3 \cdot 0 + 4 + 3 \cdot 1 + 3 + 3 \cdot 3 + 1 + 3 \cdot 0 + 2 + 3 \cdot 1 + 6 + 3 \cdot 4 + 1 \equiv 0 \pmod{10}$
 $0 + 4 + 3 + 3 + 9 + 1 + 0 + 2 + 3 + 6 + 12 + 1 = 44 \equiv 4 \not\equiv 0 \pmod{10}$
Hence, 041331021641 is not a valid UPC.

Cryptography

Section 4.6

Section Summary₆

Classical Cryptography

Cryptosystems

Public Key Cryptography

RSA Cryptosystem

Cryptographic Protocols

Primitive Roots and Discrete Logarithms

Caesar Cipher₁



Julius Caesar created secret messages by shifting each letter three letters forward in the alphabet (sending the last three letters to the first three letters.) For example, the letter B is replaced by E and the letter X is replaced by A. This process of making a message secret is an example of *encryption*.

Here is how the encryption process works:

- Replace each letter by an integer from $\mathbf{Z}_{26}$, that is an integer from 0 to 25 representing one less than its position in the alphabet.
- The encryption function is $f(p) = (p + 3) \bmod 26$. It replaces each integer p in the set $\{0, 1, 2, \dots, 25\}$ by $f(p)$ in the set $\{0, 1, 2, \dots, 25\}$.
- Replace each integer p by the letter with the position $p + 1$ in the alphabet.

Example: Encrypt the message “MEET YOU IN THE PARK” using the Caesar cipher.

Solution: 12 4 4 19 24 14 20 8 13 19 7 4 15 0 17 10.

Now replace each of these numbers p by $f(p) = (p + 3) \bmod 26$.

15 7 7 22 1 17 23 11 16 22 10 7 18 3 20 13.

Translating the numbers back to letters produces the encrypted message

“PHHW BRX LQ WKH SDUN.”

Caesar Cipher₂

To recover the original message, use $f^{-1}(p) = (p-3) \bmod 26$. So, each letter in the coded message is shifted back three letters in the alphabet, with the first three letters sent to the last three letters. This process of recovering the original message from the encrypted message is called *decryption*.

The Caesar cipher is one of a family of ciphers called *shift ciphers*. Letters can be shifted by an integer k , with 3 being just one possibility. The encryption function is

$$f(p) = (p + k) \bmod 26$$

and the decryption function is

$$f^{-1}(p) = (p - k) \bmod 26$$

The integer k is called a *key*.

Shift Cipher₁

Example 1: Encrypt the message “STOP GLOBAL WARMING” using the shift cipher with $k = 11$.

Solution: Replace each letter with the corresponding element of $\mathbf{Z}_{26}$.

18 19 14 15 6 11 14 1 0 11 22 0 17 12 8 13 6.

Apply the shift $f(p) = (p + 11) \bmod 26$, yielding

3 4 25 0 17 22 25 12 11 22 7 11 2 23 19 24 17.

Translating the numbers back to letters produces the ciphertext

“DEZA RWZMLW HLCXTYR.”

Shift Cipher₂

Example 2: Decrypt the message “LEWLYPLUJL PZ H NYLHA ALHJOLY” that was encrypted using the shift cipher with $k = 7$.

Solution: Replace each letter with the corresponding element of $\mathbb{Z}_{26}$.

11 4 22 11 24 15 11 20 9 1 15 25 7 13 24 11 7 0 0 11 7 9 14
11 24.

Shift each of the numbers by $-k = -7$ modulo 26, yielding

4 23 15 4 17 8 4 13 2 4 8 18 0 6 17 4 0 19 19 4 0 2 7 4 17.

Translating the numbers back to letters produces the decrypted message

“EXPERIENCE IS A GREAT TEACHER.”

Affine Ciphers

Shift ciphers are a special case of *affine ciphers* which use functions of the form

$$f(p) = (ap + b) \bmod 26,$$

where a and b are integers, chosen so that f is a bijection.

The function is a bijection if and only if $\gcd(a, 26) = 1$.

Example: What letter replaces the letter K when the function $f(p) = (7p + 3) \bmod 26$ is used for encryption.

Solution: Since 10 represents K, $f(10) = (7 \cdot 10 + 3) \bmod 26 = 21$, which is then replaced by V.

To decrypt a message encrypted by a shift cipher, the congruence $c \equiv ap + b \pmod{26}$ needs to be solved for p .

- Subtract b from both sides to obtain $c - b \equiv ap \pmod{26}$.
- Multiply both sides by the inverse of a modulo 26, which exists since $\gcd(a, 26) = 1$.
- $\bar{a}(c - b) \equiv \bar{a}ap \pmod{26}$, which simplifies to $\bar{a}(c - b) \equiv p \pmod{26}$.
- $p \equiv \bar{a}(c - b) \pmod{26}$ is used to determine p in $\mathbf{Z}_{26}$.

Cryptanalysis of Affine Ciphers

The process of recovering plaintext from ciphertext without knowledge both of the encryption method and the key is known as *cryptanalysis* or *breaking codes*.

An important tool for cryptanalyzing ciphertext produced with a affine ciphers is the relative frequencies of letters. The nine most common letters in the English texts are E 13%, T 9%, A 8%, O 8%, I 7%, N 7%, S 7%, H 6%, and R 6%.

To analyze ciphertext:

- Find the frequency of the letters in the ciphertext.
- Hypothesize that the most frequent letter is produced by encrypting E.
- If the value of the shift from E to the most frequent letter is k , shift the ciphertext by $-k$ and see if it makes sense.
- If not, try T as a hypothesis and continue.

Example: We intercepted the message “ZNK KGXRE HOXJ MKZY ZNK CUXS” that we know was produced by a shift cipher. Let’s try to cryptanalyze.

Solution: The most common letter in the ciphertext is K. So perhaps the letters were shifted by 6 since this would then map E to K. Shifting the entire message by -6 gives us “THE EARLY BIRD GETS THE WORM.”

Block Ciphers₁

Ciphers that replace each letter of the alphabet by another letter are called *character* or *monoalphabetic* ciphers. 字符或单字母密码。

They are vulnerable to cryptanalysis based on letter frequency.

Block ciphers 分组密码 avoid this problem, by replacing blocks of letters with other blocks of letters.

A simple type of block cipher is called the *transposition cipher* 换位密码. The key is a permutation σ of the set $\{1, 2, \dots, m\}$, where m is an integer, that is a one-to-one function from $\{1, 2, \dots, m\}$ to itself.

To encrypt a message, split the letters into blocks of size m , adding additional letters to fill out the final block. We encrypt $p_1, p_2, \dots, p_m$ as $c_1, c_2, \dots, c_m = p_{\sigma(1)}, p_{\sigma(2)}, \dots, p_{\sigma(m)}$.

To decrypt the $c_1, c_2, \dots, c_m$ transpose the letters using the inverse permutation σ^{-1} .

Block Ciphers₂

Example: Using the transposition cipher based on the permutation σ of the set $\{1,2,3,4\}$ with $\sigma(1) = 3$, $\sigma(2) = 1$, $\sigma(3) = 4$, $\sigma(4) = 2$,

- a. Encrypt the plaintext PIRATE ATTACK
- b. Decrypt the ciphertext message SWUE TRAEOEHS, which was encrypted using the same cipher.

Solution:

- a. Split into four blocks PIRA TEAT TACK.
Apply the permutation σ giving IAPR ET TA AKTC.
- b. σ^{-1} : $\sigma^{-1}(1) = 2$, $\sigma^{-1}(2) = 4$, $\sigma^{-1}(3) = 1$, $\sigma^{-1}(4) = 3$.

Apply the permutation σ^{-1} giving USEW ATER HOSE.

Split into words to obtain USE WATER HOSE.

Cryptosystems₁

Definition: A *cryptosystem* is a five-tuple (P, C, K, E, D) , where

- P is the set of plaintext strings, 明文字符串
- C is the set of ciphertext strings, 密文串
- K is the *keyspace* (set of all possible keys),
- E is the set of encryption functions, and 加密函数集
- D is the set of decryption functions. 解密函数集

The encryption function in E corresponding to the key k is denoted by E_k and the decryption function in D that decrypts cipher text encrypted using E_k is denoted by D_k . Therefore:

$$D_k(E_k(p)) = p, \text{ for all plaintext strings } p.$$

Cryptosystems₂

Example: Describe the family of shift ciphers as a cryptosystem.

Solution: Assume the messages are strings consisting of elements in $\mathbf{Z}_{26}$.

- P is the set of strings of elements in $\mathbf{Z}_{26}$,
- C is the set of strings of elements in $\mathbf{Z}_{26}$,
- $K = \mathbf{Z}_{26}$,
- E consists of functions of the form
 $E_k(p) = (p + k) \bmod 26$, and
- D is the same as E where $D_k(p) = (p - k) \bmod 26$.

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